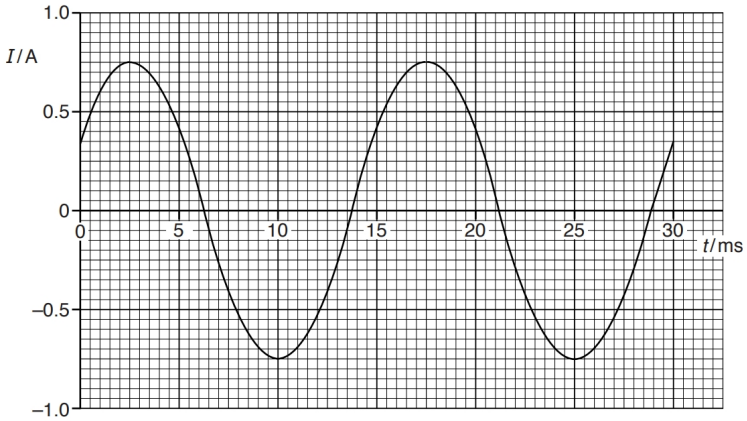
21.1 Exercises - Characteristics of alternating currents

1. The variation with time t of the sinusoidal current I in a resistor of resistance 450 Ω is shown in Fig. Use data to determine, for the time t = 0 to t = 30 ms,



a. the frequency of the current,

frequency = ... Hz [2]

b. the mean current,

mean current = ... A [1]

c. the root-mean-square (r.m.s.) current,

r.m.s. current = ... A [2]

d. the energy dissipated by the resistor.

energy = ... J [2]

2.

a. The output of a power supply is represented by:

$$V = 9.0sin20t$$

where V is the potential difference in volts and t is the time in seconds.

Determine, for the output of the supply:

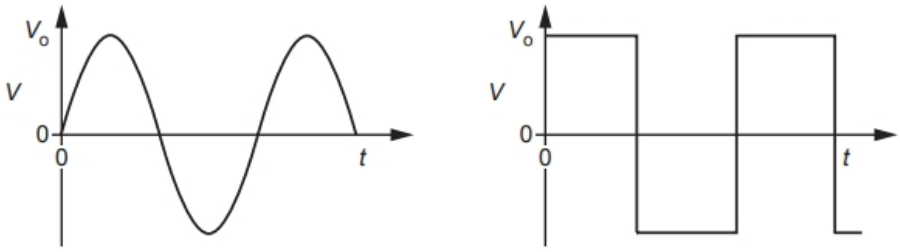
i. the root-mean-square (r.m.s.) voltage, $V_{r.m.s.}$

$V_{r.m.s.}$ = ... V [1]

ii. the period T.

T = ... s [2]

b. The variations with time t of the output potential difference V from two different power supplies are shown in Figs.



The graphs are drawn to the same scale. State and explain whether the same power would be dissipated in a 1.0 Ω resistor connected to each power supply.

.....[1]

c.

i. The power supply in (a) is connected to a transformer. The input power to the transformer is 80W.

The secondary coil is connected to a resistor. The r.m.s. voltage across the resistor is 120 V. The r.m.s. current in the secondary coil is 0.64 A.

Calculate the efficiency of the transformer.

efficiency = ... [3]

ii. State one reason why the transformer is not 100% efficient.

.....[1]

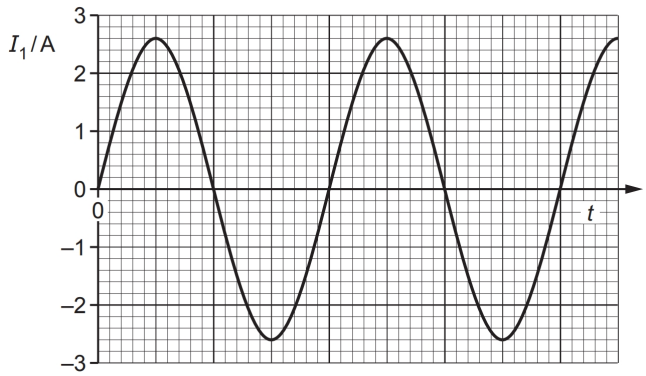
3.

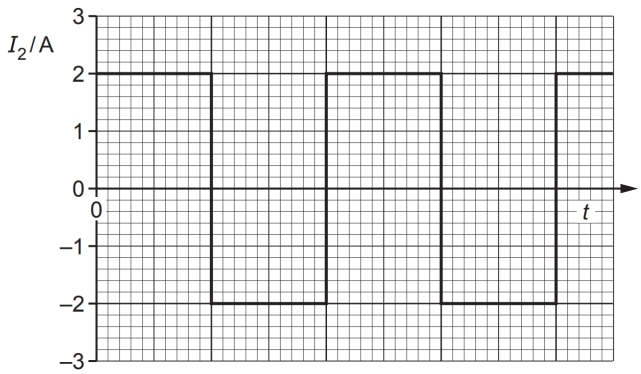
a. By reference to heating effect, explain what is meant by the root-mean-square (r.m.s.) value of an alternating current.

.....

.....[2]

b. The variations with time t of two currents I1 and I2 are shown in Figs.





i. Use Fig. to determine the peak value and the r.m.s. value of the current I_1.

peak value = ... A [1]

r.m.s value = ... A [1]

ii. Use Fig. to determine the peak value and the r.m.s. value of the current I_2.

peak value = ... A [1]

r.m.s value = ... A [1]

c. The variation with time t of the supply voltage V to a house is given by the expression

$$V = 240 sinkt$$

where V is in volts, t is in seconds and k is a constant with unit rad s^{-1}.

i. The frequency of the supply voltage is 50 Hz.

Determine k to two significant figures.

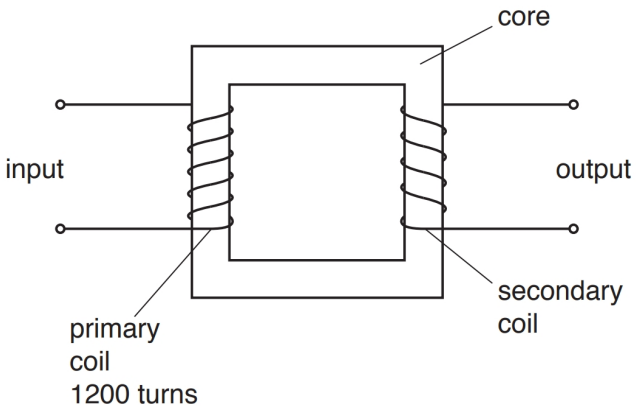
k = ... rad s^{-1} [2]

ii. The supply voltage is applied to a heater. The mean power of the heater is 3.2 kW.

Calculate the resistance of the heater.

resistance = ... Ω [2]

4. An ideal transformer is shown in Fig.



a. Explain

i. why the core is made of iron,

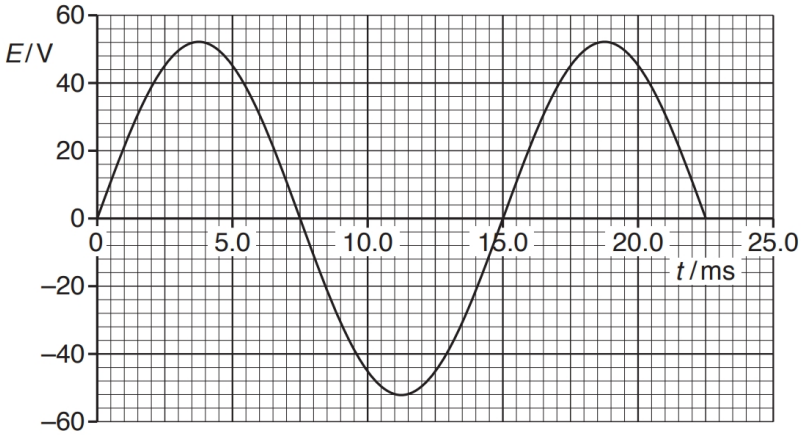
.....[1]

ii. why an electromotive force (e.m.f.) is not induced at the output when a constant direct voltage is at the input.

.....

.....[2]

b. An alternating voltage of peak value 150V is applied across the 1200 turns of the primary coil. The variation with time t of the e.m.f. E induced across the secondary coil is shown in Fig.



Use data from Fig. To

i. calculate the number of turns of the secondary coil,

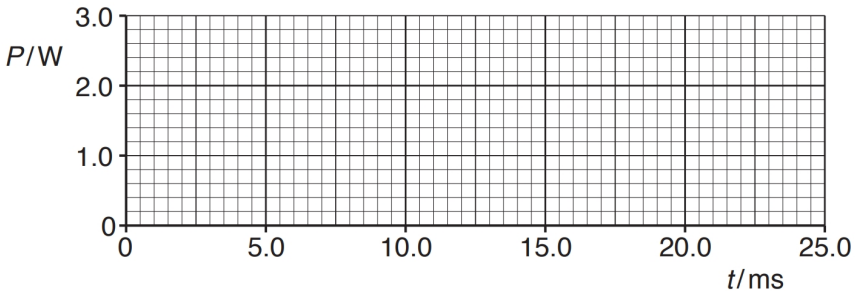
number = ... [2]

ii. state one time when the magnetic flux linking the secondary coil is a maximum.

time = ...ms [1]

c. A resistor is connected between the output terminals of the secondary coil. The mean power dissipated in the resistor is 1.2 W. It may be assumed that the varying voltage across the resistor is equal to the varying e.m.f. E shown in Fig.

i. Calculate the resistance of the resistor.

resistance = ... Ω [2]

ii. Sketch the variation with time t of the power P dissipated in the resistor for t = 0 to t = 22.5ms.